



## Genomic Medicine Programme

### MRes in Genomic Medicine – Student Handbook

ACY 2026-27



## Contents

|                                                                |    |
|----------------------------------------------------------------|----|
| Who is this handbook for? .....                                | 4  |
| Welcome .....                                                  | 5  |
| Learning Environment and Teaching Locations .....              | 6  |
| Programme Aims .....                                           | 7  |
| Learning Outcomes.....                                         | 8  |
| Teaching Methods .....                                         | 9  |
| Study Hours.....                                               | 9  |
| Attendance.....                                                | 9  |
| Module Dates and Deadlines.....                                | 10 |
| Research Project Dates and Deadlines .....                     | 10 |
| University of Cambridge Term Dates .....                       | 10 |
| Student Feedback.....                                          | 11 |
| Academic Course Leads .....                                    | 11 |
| Course Administration Team.....                                | 11 |
| Module Leads .....                                             | 11 |
| MRes – Programme Structure .....                               | 12 |
| University Access, Email and Student Card .....                | 13 |
| University Policies.....                                       | 14 |
| Student Welfare .....                                          | 14 |
| Assessment and Examination.....                                | 15 |
| Marking Scheme for Postgraduate Awards.....                    | 15 |
| Marking of Module Assignments. ....                            | 17 |
| Marking of MRes Research Projects .....                        | 17 |
| Moderation of Marks .....                                      | 17 |
| Extension to Assignment Submission .....                       | 18 |
| Students with University SSDs (Student Support Documents)..... | 20 |
| Presentation of Assignments .....                              | 20 |
| Good Academic Practice .....                                   | 21 |
| Penalties .....                                                | 23 |
| MRes Supervision .....                                         | 25 |
| MRes Research Project .....                                    | 26 |
| College Induction.....                                         | 26 |
| College Tutors.....                                            | 27 |



|                                                                                           |    |
|-------------------------------------------------------------------------------------------|----|
| Graduation .....                                                                          | 27 |
| Student Concerns and Complaints .....                                                     | 27 |
| Systems and IT .....                                                                      | 28 |
| Getting Started – Links, Guides and Training for IT and Computing .....                   | 29 |
| University of Cambridge Resources .....                                                   | 29 |
| Continuation to PhD .....                                                                 | 31 |
| Travel Around Cambridge .....                                                             | 32 |
| GM1 – Fundamentals of Human Genetics and Genomics .....                                   | 33 |
| GM2 – Research and Statistical Skills .....                                               | 35 |
| GM5 – Bioinformatics, Interpretation and Data Quality Assurance in Genomic Analysis ..... | 37 |

### Who is this handbook for?

The Genomic Medicine Programme at the University of Cambridge delivers the [NHS-commissioned Genomic Education framework](#) in partnership between the department of Professional and Continuing Education (PACE) and the School of Clinical Medicine (SCM).

The School of Clinical Medicine (SCM) admits full-time students onto the programme, studying for their MRes or MRes in Genomic Medicine. The department of Professional and Continuing Education (PACE) admits part-time students onto the programme, who study Genomic Medicine alongside their work or training.

One of the key benefits of the programme is this mixing of students from many different professional and academic backgrounds.

This handbook supports and governs all full-time, MRes students admitted onto the University of Cambridge Genomic Medicine programme by the Department of Genomic Medicine and School of Clinical Medicine (SCM).

You should follow the guidelines and adhere to the requirements set down by the Department, SCM and the wider University. Besides this handbook, further guidance will be given at various points during the course. The Genomic Medicine homepage on the Virtual Learning Environment (VLE) will contain the most up-to-date version of all documents.

If you have any questions regarding the content of this handbook, please contact your Student Administrator for the MRes in Genomic Medicine – Mr Daniel Hadow – [dh738@cam.ac.uk](mailto:dh738@cam.ac.uk).

*Whilst every effort is made to avoid changes to this programme, published details may be changed without notice at any time.*

*The Department of Professional and Continuing Education (PACE) and the School of Clinical Medicine (SCM) reserve the right to withdraw or amend any part of this programme without prior notice.*

## Welcome

We are pleased to welcome you to the University of Cambridge and onto the Genomic Medicine programme.

This course was originally designed in response to a request from Health Education England (HEE), the governmental body responsible for the coordination of education and training in the healthcare sector in England. They wished to promote courses designed for suitably qualified NHS healthcare professionals from across multi-professional teams in bring up-to-date knowledge of genomic techniques and practices to healthcare, and to exploit the UK's established expertise in this field.

Lectures will be co-delivered with students from all courses within the Genomic Medicine programme at Cambridge, including part-time students with Postgraduate Certificate (PGCert), Postgraduate Diploma (PGDip), Master of Studies (MSt), for credit and CPPD only modular students, as well as full-time students studying for the Master of Philosophy (MPhil) and the Master of Research (MRes). The courses attract students from many different professional and academic backgrounds. We hope that you will be able to take advantage of the different perspectives, experience, research and academic expertise that such as diverse group brings to this programme.

The University of Cambridge is one of the oldest and most prestigious academic institutions in the world. Cambridge itself is a unique place to live and study. Moreover, the Biomedical Campus is a vibrant and ever-expanding place where pioneering biomedical research takes place.

As well as the formal contents of the course, there are talks, lectures and seminars that are open to you as a member of the University. For Masters level students, College life also provides a wide variety of academic, sporting and leisure activities for you to take advantage of. We encourage you to make the most of these opportunities.

We aim to ensure that the course runs as smoothly as possible, and that it is a positive experience for staff and student alike. If you do encounter any problems with the course arrangements, the taught material, or with your research project, please raise them as soon as possible, giving us the best opportunity to address them.

Enjoy your time at Cambridge, and we look forward to meeting you.

Prof Marc Tischkowitz, Programme Director & Head of Department  
Professor of Medical Genetics and Genomic Medicine, Department of Medical Genetics

Dr Timothy Hearn, Course Director – MPhil/MRes in Genomic Medicine.

## Learning Environment and Teaching Locations

The University of Cambridge is the UK's leading research university. In the 2021 Research Excellence Framework exercise, the University's School of Clinical Medicine (SCM) was ranked first in the UK for internationally recognised research. In partnership with associated hospitals, the University is one of only eight UK Academic Health Science Centres. This prized status is awarded by an international panel in recognition of the University of Cambridge as an internationally leading University Medical Centre, with groups that are globally competitive in translating excellence in biomedical science into effective healthcare.

Located on the southern edge of Cambridge, the Cambridge Biomedical Campus contains the [University of Cambridge School of Clinical Medicine](#) (SCM) with its various departments and associated institutions, [Addenbrookes Hospital](#), a major University Teaching Hospital WITH 1100 beds and regional services to the whole of the East of England, and a directly funded Research Institute – the [MRC Laboratory for Molecular Biology](#) (LMB). In addition to the LMB there are four further MRC units located on the campus.

The University's main Biological Science campus is located on the Downing Site in the centre of Cambridge and houses the [School of Biological Sciences](#), which has particular strengths in developmental, cellular and molecular research. Research within the School of Biological Sciences pursues several themes that intersect with research themes in the School of Clinical Medicine. Here we utilise the state of the art bioinformatics training facilities available in the [Craik-Marshall building](#) and the [Pembroke Teaching Rooms](#).

The [Department of Genomic Medicine](#) is based on Level 6 of the Addenbrookes Treatment Centre on the Biomedical Campus. The majority of teaching takes place in the School of Clinical Medicine building on the Biomedical Campus.

To the south of Cambridge are the [Brabham Institute](#) and the Hinxton Genome Campus with the [Wellcome Sanger Institute](#) and [EMBL-European Bioinformatics Institute](#).

This location in Cambridge of a major regional University Hospital, Medical School, and Research Institutes on a single campus, together with surrounding strengths in Biological Science within a ten-mile radius is unusually within the UK and offers an exceptional environment for highest quality collaborative biomedical research, and its translation into healthcare.



## Programme Aims

The MRes in Genomic Medicine addresses the perceived skills deficit in data science and computational skills that is common in many postgraduate students. The MRes in Genomic Medicine allows students to complete a research focussed degree incorporating a large element of data science-related teaching. The MRes will first equip research students with the skills necessary to undertake computational genomics projects through a taught programme. Students then utilise these skills to plan and complete a research project.

- Provide a research environment where students can learn the practical skills required for performing research in Genomic Medicine.
- Prepare and upskill students with the most up to date computational and bioinformatic knowledge for performing genomic research.
- Develop students independent research skills and understanding of research methodologies and clinical opportunities relevant to genomic medicine.
- Develop researchers' competence in using genomic and data science technologies for biomedical research.
- Prepare students for either higher study (PhD/MD) or career roles in a research-intensive environment in genomic medicine-related fields.
- Develop a cohort of students who are proficient in the wide variety of research skills necessary to plan and deliver Genomic Medicine research projects.
- Encourage a commitment to intellectual challenge and evidence-based clinical practice informed by the latest conceptual and theoretical knowledge of genomic medicine.
- Develop students' intellectual, practical and transferable skills related to genomic medicine.
- Encourage critical thinking related to genomic medicine.
- Develop good scientific practice and record keeping.
- Increase employability in the local genomics industry centred around Cambridge, where data science skills are highly valued.
- Provide professionally relevant teaching and learning informed by research in an integrated clinical and research environment.
- Conduct systematic literature reviews relevant to their research areas.
- Equip students for entry into health care professional training schemes including postgraduate entry medicine courses.
- Provide training for clinician scientists who wish to work on data produced from Genomics England and the NHS Genomic Medicine service as part of the diagnostic and treatment pathway.



## Learning Outcomes

- To enhance the students' knowledge and critical understanding of recent developments in genomic medicine.
- To develop students' knowledge in genomics research skills relevant to variant interpretation, rare disease, cancer and basic biomedical science.
- To develop students' knowledge and understanding of genomic medicine informed by research in a rapidly changing integrated clinical and research environment.
- To enable deployment of new knowledge in a research environment.
- To develop an understanding of genomic technologies and where genomic information is used as part of the diagnostic and treatment pathway.
- To develop students' knowledge so that they have the confidence to propose and design high quality research.
- To update and extend students' understanding of research methodologies and clinical opportunities.
- To demonstrate knowledge, abilities and skills to engage in focused, professionally relevant, independent learning, and through the production of a dissertation.
- Students will be taught an up-to-date basic toolkit of skills needed for Genomic Medicine as a data science including the use of the Unix command line, R and Galaxy.
- Students will gain practical experience in the central computational techniques in genomics; WGS, WES, variant interpretation, and RNA-seq.
- Students will be taught the skills to access genomic tools and datasets from the EBI and Genomics England.
- Students will gain understanding of experimental techniques for variant interpretation, including PCR, qRT-PCR, and techniques for exploring splice variants.
- Students will gain understanding of the experimental techniques for NGS library preparation.
- Students will develop the skills necessary to locate, read, interpret and analyse primary and secondary sources of material enabling the development of a conceptual and theoretical understanding of recent developments in genomic medicine.
- Students will enhance their skills to evaluate current scholarship and research critically and to place this knowledge within the context of their own situation and practice as clinical leaders.
- Students will develop the ability to formulate a research topic relevant to their clinical context, to collect and analyse primary and/or secondary sources of data, and to undertake professionally relevant research.
- Students will enhance the facility to communicate the results of their ideas, research and its conclusions in a written form acceptable as a work of scholarship potentially publishable in a professional or academic journal.

## Teaching Methods

Teaching takes place during week-long modules, approximately 9-5, Monday-Friday (*with the exception of GM2*). In person attendance is mandatory, it is not possible to attend modules remotely.

Teaching methods will include, but are not limited to lectures, presentations by guest speakers and facilitators, interactive learning activities, reading and assignments to be completed by students outside of classroom sessions.

- Teaching methods will include a combination of interactive seminars, large and small groups work with emphasis on peer review and reflection,
- Face-to-face delivery, with syllabuses and reading lists given to students, online support between modules and feedback on all assignments,
- Seminars, discussions and lectures led by a tutor with specialist knowledge of the content,
- Guest lectures and seminars presented by leaders in their field,
- Personal study guided by appropriate syllabuses, reading lists and resources provided,
- Full and appropriate use of the programme's Virtual Learning Environment (VLE),
- Small group in person supervisions with your programme supervisor.

## Study Hours

Each module runs for one week (approximately 9-5, Monday-Friday). With the removal of breaks, each module equates to approximately 30 hours of in person teaching. It is expected that on top of the 30-hours teaching, students will attend an in-person supervision following the completion of each module, read widely around the subject matter delivered in each module and complete each module assignment.

The award of academic credit is a means of quantifying and recognising learning within the UK. One credit notionally represents 10-hours of learning. Each module is worth 15 credits, so students should expect to study for approximately 150 hours to complete each module successfully. However, it is recognised that student's study at different paces and use a variety of approaches, so this is a recommendation, rather than a hard-and-fast calculation.

## Attendance

Each module will consist of five days of teaching. Full attendance is expected, with teaching dates made available well in advance. A register for each module will be maintained and attendance recorded. If you are unable to attend during a module, due to illness or a prior appointment, you should contact your course administrator – Mr Daniel Haddow – [dh738@cam.ac.uk](mailto:dh738@cam.ac.uk).

Persistent non-attendance will be addressed by the Course Director.

## Module Dates and Deadlines

The key dates for all modules in the 2025-26 academic year are summarised below. Detailed information, including teaching rooms and timetables, will be communicated to enrolled students prior to each module via the Virtual Learning Environment (VLE).

Whilst every effort is made to deliver teaching at the published locations as stated below, locations are occasionally subject to change.

| Module                                                                                   | All-day teaching                                                        | In-person at                                                 | Assignment due                           |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------|
| <b>GM1 Fundamentals of human genetics and genomics*</b>                                  | 12 – 16 Oct 2026                                                        | School of Clinical Medicine<br>Hills Road, Cambridge CB2 0QQ | Mo 16 Nov<br>2026<br>@12 noon UK<br>time |
| <b>GM2 Research and statistical skills for genomic medicine</b>                          | <b>20, 23, 29 Oct 2026<br/>5, 19, 26 Nov 2026<br/>10 Dec 2026 (opt)</b> | School of Clinical Medicine<br>Hills Road, Cambridge CB2 0QQ | Mo 18 Jan<br>2027<br>@12 noon UK<br>time |
| <b>GM5 Bioinformatics, interpretation and data quality assurance in genomic analysis</b> | 30 Nov – 4 Dec 2026                                                     | School of Clinical Medicine<br>Hills Road, Cambridge CB2 0QQ | Mo 11 Jan<br>2027<br>@12 noon UK<br>time |

## Research Project Dates and Deadlines

Project Proposal Deadline – **Friday 11 December 2026**

Project Submission Deadline – **Friday 27 August 2027**

Project Viva Dates – **First two weeks of September 2027 (exact dates tbc)**

## University of Cambridge Term Dates

Not all taught modules run during term time; however, the University terms are relevant to several University processes and procedures.

**Michaelmas Term** – Tuesday 6 October 2026 to Friday 4 December 2026

**Lent Term** – Tuesday 19 January 2027 to Friday 19 March 2027

**Easter Term** – Tuesday 27 April 2027 to Friday 18 June 2027

Please also see: [Term dates | University of Cambridge](#)

## Student Feedback

Your feedback is essential for the continued development and evaluation of the Genomic Medicine at the University of Cambridge. As part of your course, you will be asked to provide feedback at the end of each taught module. Please do provide your feedback as it is also important to us in developing and continuously improving the course and to your course tutors and lecturers as part of their own professional development. We aim to provide you with an update on the VLE after each module survey has closed to report back on how your feedback will be acted upon. We also ask you to complete an overall questionnaire at the end of the course. We are then able to reflect on your comments and implement changes as necessary to improve your experience and the experience of subsequent cohorts.

## Academic Course Leads

Contacts for questions about course content, academic support and concerns.

|                              |                                                                                                                           |                                                      |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Dr Timothy Hearn</b>      | <b>Genomic Medicine<br/>Course Director – MPhil/MRes</b><br>University of Cambridge School of Clinical Medicine           | <a href="mailto:tjh70@cam.ac.uk">tjh70@cam.ac.uk</a> |
| <b>Prof Marc Tischkowitz</b> | <b>Programme Director<br/>Head of Department, Genomic Medicine</b><br>University of Cambridge School of Clinical Medicine | <a href="mailto:mdt33@cam.ac.uk">mdt33@cam.ac.uk</a> |

## Course Administration Team

Contact for general and technical questions, procedures, requests, advice and support. Many answers can be found in this handbook.

|                         |                                                                                     |                                                      |
|-------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Mr Daniel Haddow</b> | <b>Genomic Medicine<br/>Student Administrator</b><br>Department of Genomic Medicine | <a href="mailto:dh738@cam.ac.uk">dh738@cam.ac.uk</a> |
|-------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------|

## Module Leads

Contacts for module specific academic questions.

|                         |                                                                                  |                                                                                        |
|-------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Dr Gemma Chandratillake | GM1 Introduction to human genetics and genomics                                  | <a href="mailto:Gemma.chandratillake@nhs.net">Gemma.chandratillake@nhs.net</a>         |
| Dr Tim Hearn            | GM2 Research and statistical skills for genomic medicine                         | <a href="mailto:tjh@cam.ac.uk">tjh@cam.ac.uk</a>                                       |
| Dr Leonardo Bottolo     | GM2 Research and statistical skills for genomic medicine                         | <a href="mailto:lb664@cam.ac.uk">lb664@cam.ac.uk</a>                                   |
| Dr Tim Hearn            | GM5 Bioinformatics, interpretation and data quality assurance in genome analysis | <a href="mailto:tjh@cam.ac.uk">tjh@cam.ac.uk</a>                                       |
| Dr Kenneth Langlands    | GM5 Bioinformatics, interpretation and data quality assurance in genome analysis | <a href="mailto:Kenneth.langlands@pace.cam.ac.uk">Kenneth.langlands@pace.cam.ac.uk</a> |

## MRes – Programme Structure

The MRes in Genomic Medicine is a 12-month programme, administered and owned by the Department of Genomic Medicine, part of the School of Clinical Medicine (SCM). The MRes is split between taught and research elements. The research element is worth 70% of your degree, whilst the taught element is worth 30%.

The course is a full-time component of the University of Cambridge Genomic Medicine programme. The degree awarded is an MRes (by advanced study). **To complete the MRes, students must complete three mandatory taught modules (see module dates and deadlines on page 9) and a research project.**

The passing mark for each module assignment and dissertation is 60%.

- 60-74% = Pass
- 75%+ = Pass with Distinction

For more information see the marking scheme for postgraduate awards on page.

Each student is assigned a programme supervisor; this is usually your Course Director. Your programme supervisor will provide appropriate academic support throughout the course. You will be invited to attend regular, small-group supervisions with your programme supervisor throughout the year.

Each student will be assigned a research supervisor, once research proposals have been submitted in February. Your research supervisor will be an expert in the field of your specific research. They will provide specialised support in the completion of your project. It is your responsibility to arrange meetings with your research supervisor throughout the duration of your project.



[University Access, Email and Student Card.](#)

### **Student Registration Exercise**

All students enrolled on an award bearing course are required to complete an annual registration exercise for the UK Higher Education Statistics Agency (HESA). This involves confirming and/or updating the personal details we hold about you. It is completed online and only takes a few minutes. At the same time, you are asked your preferred method of contact and who to contact in case of emergency.

On the final screen of the Student Registration Exercise, you be able to generate and collect your University of Cambridge username, known as your CRSid, and your initial password to log in to get your student account and email. Your email address will be in the form of [yourCRSid]@cam.ac.uk and will allow you access to Office365, including OneDrive. For more information on how to access your CRDId, see the [University Information Services](#) (UIS) website.

For more information on the Student Registration Exercise, see [Student registration | Cambridge students](#)

*A link to the Student Registration Exercise will be sent to you by the Student Registry closer to the start of the course.*

### **University Email Address**

Please note that all communications relating your course will be done via your @cam email address. There will be an initial 2-week grace period at the start of your course to allow to complete the Student Registration Exercise and set up your @cam email account.

### **University Card**

As you are studying a full-time programme you will be given a University Card. The University Card grants access to university premises such as the Central Library.

Your college will produce your University Card in the run up to the start of Michaelmas Term. Your college will contact you via your new @cam email address to arrange the pick up of your University Card from your College's Ports Lodge. If you lose your card, or it expires, please contact your college.

### **University Account Access**

Your university account – formerly known as your RAVEN account, is the University's central web authentication service. For more information click [here](#). You can collect your password on successful completion of the Student Registration Exercise.

### **Virtual Learning Environment (VLE)**

The VLE is central to your studies on the Genomic Medicine programme. Through the VLE you will access important course information throughout your studies. You will be enrolled into the course space for each, individual module.



You will be enrolled onto the VLE using your new @cam email address. The PACE e-learning team will contact you to confirm your account and prompt you to set up a password. If you have any issues with access, or any wider questions regarding the VLE, contact your Course Administrator.

The VLE login page is here - [Log in to the site | PACE VLE](#).

You will also be enrolled into the Central University Moodle space - [Log in to the site | Moodle](#). Here you will find information regarding supervisions and further information around your research project.

## University Policies

Policies that all students should familiarise themselves with include:

[Student Discipline](#)

[Freedom of speech](#)

[Public gatherings](#)

There are a number of legal regulations that all staff and students are obliged to comply with:

[Disclosure and Barring Service \(DBS\) checks](#)

[Copyright Compliance](#)

[Information Compliance](#)

[Data Protection Act \(including details about how we use your personal information\)](#)

[Freedom of Information Act](#)

More information on the items above can be found at:

<http://www.cambridgestudents.cam.ac.uk/new-students/rules-and-legal-compliance>

## Student Welfare

The welfare of students is of the utmost importance to us at the University and there are various sources of advice and support. Your first point of contact should be your college tutor (you will be assigned a college tutor when you matriculate), your Course Director and/or your Student Administrator.

However, there may be circumstances where you wish to consult someone more independent of your daily environment. The following organisations are available:

- The [University's Accessibility and Disability Resource Centre \(ADRC\)](#)
- The [Cambridge University Students' Unions' Advice Service](#)
- [Equality, Diversity and Inclusion Service](#)



Further information on general wellbeing, mental health and graduate study can be found at [www.studentwellbeing.admin.cam.ac.uk/](http://www.studentwellbeing.admin.cam.ac.uk/) and [www.cambridgestudents.cam.ac.uk/your-course/graduate-study](http://www.cambridgestudents.cam.ac.uk/your-course/graduate-study)

## Assessment and Examination

### Assessment

Each module generally requires submission of an assignment of approximately 2000-3500 words or equivalent.

Assignments are usually due approximately 4-weeks after the end of the module. Details of the assessment methodology and assignment will be provided as part of the individual taught module. For a complete overview of all modules, with content and learning outcomes, see the module descriptions at the end of this handbook. For an overview of all assessment assignment deadlines, see page 9.

In addition to the assignment requirements of the taught modules, MRes students must also complete a research project.

### Rules of general application for assignments and dissertation.

- All assignments for examination should be submitted by the specified deadline and in the electronic version as specified in the assignment via the VLE/Moodle
- Candidates should not put their name on any assignment submitted for examination; they should instead use their allocated candidate number. *This does not apply to the research project, which is submitted as a named piece of work.*
- All coursework submitted for formal assessment and examination, including assignments and research project, will be checked by plagiarism detection software, Turnitin UK.

## Marking Scheme for Postgraduate Awards.

Under the universal mark scheme, 60% is the pass mark and a distinction is 75% or over.

See below (page 16) for the general assignment/dissertation mark scheme.



| Score                 | Mark Awarded           | Student's work shows                                                                                                                                                                                                                                                                                                |
|-----------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Excellent</b>      |                        |                                                                                                                                                                                                                                                                                                                     |
| 80-100                | Pass with distinction. | Evidence of the exceptional quality in relation to the criteria listed for the award of 70-79% and outstanding critical insights and thought- provoking arguments.                                                                                                                                                  |
| 75-79                 | Pass with distinction. | An accessible, accurate and clear account.<br>Clear assimilation and understanding of the evidence. Well informed by a wide range of relevant ideas.<br>Excellent analyses, arguments and explanations. Exceptionally good structuring of the material with clear progression and development as the work proceeds. |
| <b>Good</b>           |                        |                                                                                                                                                                                                                                                                                                                     |
| 70-74                 | Pass                   | An accessible, accurate and direct account.<br>Clear assimilation and understanding of the evidence. Well informed by current ideas.<br>Very good analyses, arguments and explanations.<br>Very good insights and personal reflections on the material. Carefully structured and organised presentation.            |
| <b>Competent</b>      |                        |                                                                                                                                                                                                                                                                                                                     |
| 65-69                 | Pass                   | An accessible, accurate and direct account. Good analyses, arguments and explanations.<br>Good insights and personal reflections on the material. Well-organised presentation.                                                                                                                                      |
| 60-64                 | Pass                   | An accessible, accurate and direct account.<br>Fair analyses, arguments and explanation but with some remaining gaps or confusion.<br>Fair degree of personal insight. Reasonably well organised presentation.                                                                                                      |
| <b>Pass threshold</b> |                        |                                                                                                                                                                                                                                                                                                                     |
| 50-59                 | Fail                   | Reliance on a restricted range of evidence or irrelevant material introduced.<br>Weaknesses of factual description.<br>Weaknesses in the analyses, arguments and explanations. Weaknesses in the insights and reflections on the material.<br>Weakly organised presentation with poor progression through the work. |
| 0-49                  | Fail                   | Limited range of evidence or lack of focus. Weak understanding of the material presented. Lack of coherent argument.                                                                                                                                                                                                |

### Marking of Module Assignments.

Module assignments are graded independently by two assessors, “Marker 1” and “Marker 2”, according to the postgraduate marking scheme shown above (page 16). This process is anonymous and there is an internal moderation process to ensure that grades are fair. Each assessor will provide written feedback via Turnitin comments on the VLE. Students can review these comments once the marks for the assignment have been released. The final mark for the assignment is an average of the marks awarded by each assessor. Individual marker grades are not shared with each student.

Where assignments are comprised of multiple sections, different assessor pairs may mark each component and give feedback via the Turnitin comments section in the VLE. In this case final marks are an average of the individual marks awarded for subsections, unless a different weighting of subsections has been indicated on the assignment prompt.

Assignment marks are released via the VLE for all students at the same time, and students will be informed of this release via email. The Course Director will contact any students receiving a fail mark (i.e. less than 60%) to discuss the implications of this with you.

**Please note that all marks are considered provisional until they have been approved by an exam board.**

### Marking of MRes Research Projects

Two examiners will mark the research project, and each will report independently. The examiners are required to assure themselves that the dissertation is clearly written, takes account of previously published work on the subject and represents a contribution to scholarship.

In addition to the marking scheme for postgraduate awards, examiners will use a research project feedback form to mark projects. Students will be sent their project feedback forms after marks have been approved by an exam board.

MRes students participate in a viva discussion with two examiners in September 2026. This can be in-person or remote. The viva is marked and contributes to the mark awarded for the research project.

**Resubmissions are not permitted for students in the Genomic Medicine Programme.**

### Moderation of Marks

Please note that all marks posted on the VLE are not considered final until they have been moderated in a meeting of internal and external examiners. Moderation takes place in October after the end of the academic year. If a student has any outstanding unmarked assignments (due to extensions) and cannot be included in the moderation meeting, this students' confirmation of their award is delayed until their grades can be moderated. This may be as late as the moderation meeting at the end of the next academic year.

## Extension to Assignment Submission

### Short extensions (maximum 3 weeks)

The deadline for the submission of a course assignment will always be clearly signalled to you in writing and will normally be in the course guide, published before the start of the course. All submission deadlines refer to UK Time. Check carefully if your local time differs from this.

It is important to manage your assignment deadlines, so you can submit your work on time. Your Course Director and Supervisors will be able to provide techniques on how to manage your deadlines and there are useful study tips and advice on time management in the Student Help section on the VLE. However, we understand that sometimes things happen which may affect your ability to study or prevent you from submitting your work. If this happens in the time leading up to your assignment submission you are advised to contact your Course Administration team to discuss your options.

The type of extensions you can request are:

1. Up to 7-days self-certified extension, no evidence required (one per academic year, not available for research project submission deadlines)
2. 7 days self-certified extension, followed by 2 weeks certified extension, evidence required.
3. Up to 3 weeks certified extension, evidence required

If you request a 7-days self-certified extension, you do not need to provide evidence to support your request nor are you required to provide a specific reason. However, it is essential that you let the course team and Supervisor/Course Director know prior to or on the submission deadline. Please be aware that this type of extension can only be requested once per academic year and is not available in relation to the research project.

Certified extensions will require the submission of evidence in support of your application. Please ensure you discuss your options with the course team and your tutor to fully understand the repercussions on subsequent deadlines, workload, marking or being able to complete your studies in time.

Examples of grounds for a short-term extension request:

- Short-term illness of self or dependants;
- Unanticipated changes in personal circumstances;
- Unexpectedly heavy work demands.

The following reasons would not normally be accepted as valid reasons for a short-term extension:

- Misreading information about submission deadlines;
- Holiday arrangements;
- Religious festivals;
- Social commitments;
- Technical software, hardware, network or internet problems.



To request an extension, you should complete the online [Extension Request Form](#) in the VLE before your assignment deadline and select which extension you are requesting. Your Course Administration team can provide you with a copy of the form and guide you through the process.

Regardless of the nature of your extension request, for certified extension requests you will normally be asked to provide documentary evidence to support your application. The exact evidence will depend on the nature of the claim but may include a medical letter from your GP or hospital, a letter from your employer or legal documents. Any evidence provided must be time relevant to the period you are making a claim for. If the supporting documentation is not in English, a translation must accompany the document. All supporting documentation must be submitted together with the request for an extension, or the request will be rejected.

All documentation will be treated confidentially within the department and will only be viewed by those necessary in the decision-making process. If an extension is granted, you would be expected to meet the agreed deadline.

Please be aware when requesting an extension of the repercussions on subsequent deadlines. Falling behind due to an extension is not a valid reason for requesting another extension to subsequent assignments. Please also be aware that extensions may result in not completing with your cohorts and delays in receiving ratified marks.

### **Research Project Extensions**

If you require a longer (3-weeks or more) extension to your project submission deadline you can apply “to extend your submission deadline” through your CamSIS self-service portal. Any such request should be submitted well in advance of the original submission deadline. Please note, it can take several weeks for an application to be considered via the various layers of approval.

Applications to extend the end of registration date should be made online via the CamSIS self-service: <https://www.cambridgestudents.cam.ac.uk/your-course/postgraduate-study/your-student-status/extending-your-submission-date>

Please note that extensions to your dissertation submission deadline may affect the timeliness of your marks being approved and subsequently when you will receive your award. It is likely that in the event you receive a longer extension to your dissertation deadline you will not receive your award and therefore graduate until the next academic year. This will have a knock-on effect on your ability to take up a place on another programme, for example a PhD, or access a graduate visa to stay in the UK if you are an overseas student.

**Please note that it is not possible to retrospectively apply extensions after the original submission date.**

### Students with University SSDs (Student Support Documents)

If you have (or think you may have) a disability or other longstanding circumstances that will *always* impact your ability to submit assignments on time, please contact the University's [Accessibility & Disability Resource Centre](#) at the beginning of the academic year. They will work with you to assess if they can issue a University SSD. An SSD may include instructions about modifications to marking and submission deadlines. If the Genomic Medicine course administration team receives an SSD for you, they will contact you to discuss any specifications around assessment. Students with SSDs may receive a “default” extension of two weeks to every assignment. Students with such arrangements may still follow the extension procedures outlined above if they require additional time for an individual assignment due to unforeseen circumstances.

### Presentation of Assignments

#### Word Count

It is not a requirement to write up to the word limit for your assignment. It is provided as an upper limit but also provides guidance on the number of words expected to complete an assignment thoroughly. Whilst some individuals can write in a very concise manner, you might consider whether you are missing any aspects of the assignment if your word count is much lower than the assignment limit. Figure legends, tables, references, and the question itself should not be included in the word count. The word count provided on submission should accurately reflect the word count of the submitted assignment.

#### Referencing

A reference or citation is a way of acknowledging where you made use of the work of others and the correct presentation of citations and references is an important part of any piece of academic writing. Students are expected to cite and reference correctly to avoid committing academic misconduct.

To reference and cite correctly, a recognised reference system should be used. Please see the University's page on recognised reference systems: [Referencing conventions | Plagiarism and Academic Misconduct](#)

Students on the Genomic Medicine programme can use either the [Harvard style](#) or [Vancouver style](#) in all their work for the course.

The use of a citation management tool such as EndNote, Mendeley or Zotero is strongly recommended.

#### Genetic Nomenclature

Human gene names should be capitalized and italicized e.g. *TP53*. A gene has the same name whether it includes a variant/mutation or not. Everyone/every cell has the gene (in most cases two copies of it), but not everyone/every cell has the variant/mutation. So, the phrase “driver mutations like *TP53*” is incorrect as *TP53* is a gene not a mutation/variant. It should be phrased “driver mutations like mutations in *TP53*”, similarly “She has *BRCA1*” should be phrased “She



has a pathogenic variant in *BRCA1*” since we all have *BRCA1*. Scientists will often refer to a change in a gene as a mutation, but clinically, the term “variant” is preferable, both because the pathogenicity of a variant is often not clear and the term variant carries fewer negative connotations. This site is useful if you are trying to understand variant nomenclature:

<http://varnomen.hgvs.org/bg-material/basics/>

## Figures and Tables

Figures and tables can often help make a point more clearly and careful selection of figures/tables and/or drawing of original figures/tables can demonstrate your understanding of the material. Guidelines for the inclusion of figures/tables:

- If including a figure/table from the literature, either through direct copy & pasting or by redrawing a figure heavily inspired by a figure/table from the literature, such figure/tables should be properly referenced.
- If drawing a completely original figure/table, please indicate that this is your own work since it can be difficult to distinguish original figures from un-referenced figures and we want to ensure you receive appropriate credit for your work.
- Figure legends/table text should not be included in word counts.
- Free text should not be turned into a figure/table simply to extend word count

## Presentation of the MRes Research Project

The areas which are considered in the presentation and marking of the project include: presentation of assignment; clarity of expression and logical flow; referencing; discussion of literature; use of bioinformatics theory; critical approach; appropriate use of statistics and interpretation of data; and use of figures / tables. The dissertation should contain specific sections: Background, Methods, Results, Discussion, and Conclusion.

More detailed guidance on the writing and assembly of the dissertation is contained in a separate booklet that will be provided on the dissertation VLE space.

## Good Academic Practice

University guidance on good academic practice and plagiarism, including information on study skills and referencing conventions: <https://www.plagiarism.admin.cam.ac.uk/>

## Turnitin

All assignments and dissertations will be submitted to Turnitin UK text-matching software. Turnitin compares the text of submitted work to sources in its database, which is made up of internet content, selected journals, and previous student submissions. Turnitin is only one method of checking the originality of a student’s work and examiners may use other investigative procedures if they have any queries regarding originality. See

<https://www.plagiarism.admin.cam.ac.uk/investigating/turnitin>

## Academic Misconduct

The University defines academic misconduct as the “gaining or attempting to gain, or helping others to gain or attempt to gain, an unfair academic advantage in formal University assessment, or any activity likely to undermine the integrity essential to scholarship and research. It includes being in possession of unauthorised materials or electronic devices during an examination, including recording or communication devices or devices that can store data, even where registered students are unaware that such materials or devices are unauthorised, have no intention of using them, or are unaware that they have them in their possession.”

Academic misconduct includes:

- **Plagiarism:** using someone else’s ideas, words, data, or other material produced by them without acknowledgement.
- **Self-plagiarism:** using the registered student’s own ideas, words, data or other material produced by them and submitted for formal assessment at this University or another institution, or for publication elsewhere, without acknowledgement, unless expressly permitted by the assessment.
- **Contract cheating:** contracting a third party to provide work, which is then used or submitted as part of a formal assessment as though it is the registered student’s own work.
- **Collusion:** working with others and using the ideas or words of this joint work without acknowledgment, as though it is the registered student’s own work, or allowing others to use the ideas or words of joint work without acknowledgment.
- **Impersonating someone or being impersonated** in an examination or arranging for someone to impersonate someone else by sitting their examination.
- **Fabrication, falsification or misrepresentation** of data, results or other outputs or aspects of research, including documentation and participant consent, or presenting or recording such data, etc, as if they were real; or
- **Failure to meet legal, ethical and professional obligations** in carrying out research. This includes failure to follow agreed protocol if this failure results in unreasonable risk or harm to humans, other sentient beings or the environment, and facilitating misconduct in research by collusion in, or concealment of, such actions by others. It includes any plan or conspiracy to attempt to do any of these things.

Please ensure you are familiar with the University’s Plagiarism and Academic Misconduct guidance for referencing:

<https://www.plagiarism.admin.cam.ac.uk/resources-and-support/referencing/referencing-conventions>

## Penalties

### Penalties for late submission

The following penalties apply to all individual module assignments and your final dissertation.

Where a student does not submit their assignment by midday UK time on the submission deadline day, or where an extension to the submission deadline has not been granted, the following penalties shall apply:

*Penalty of 5% deducted from the score for the piece applied for each day, or part thereof, after the deadline date. Up to a maximum of 5 days. Any late penalties incurred up to 5-days will not reduce your mark to below the pass threshold (60%).*

*More than 5 whole days after the deadline would result in a Fail mark being applied to that piece of work.*

### Penalties for word count violations

The following penalties apply to all individual module assignments and your final dissertation.

It is not required to write up to the upper word limit for your assignment/dissertation. It is provided as a guide on the number of words expected to complete an assignment thoroughly. Whilst some individuals can write in a very concise manner, you might consider whether you are missing any aspects of the assignment if your word count is much lower than the assignment range. The word count should not include appendices, bibliography, tables or figures. You will include the word count on your submission and checks will be carried out by the Course Administration Team to ensure that the declared word count is accurate.

The word count **should** include in-text citations. Students may consider using the Vancouver referencing system which minimises in-text word counts – see referencing on page 19.

The word count **should not** include appendices, bibliography, tables, figures or figure legends. However, adding long narratives into tables, figures or legends **should not** be used as a means to evade word count penalties.

Should the word count go above the upper limit allowed for the assignment/dissertation you will be contacted and asked to submit an assignment/dissertation that complies with the word limit. If this results in submitting after the deadline (and permission has not been given for an extension to the deadline) then late submission penalties apply as per those set out in 'penalties for late submission'.

*Should you not resubmit your assignment to comply with the word count, the assignment will be marked as a fail.*

*Resubmitted assignments must be resubmitted within the word count within 5-days of the original deadline (if an extension has not been granted), failure to do so will result in the assignment being marked as a fail.*

## Resubmissions

*Resubmissions are not permitted on an assignment or dissertation in the case of a fail. Instead, students are able to apply to the EAMC at the end of their course (see mitigating circumstances below).*

## Mitigating Circumstances

The mitigating circumstances process is intended to help those who have experienced severe or unexpected difficulties at the time of submitting an assignment. The effect on the submission or performance of the assignment must have been seriously adverse, unanticipated and beyond your control.

This process is applicable **only after the deadline** for the assignment submission. If you are anticipating difficulty in advance of an assignment deadline, you should contact your student administrator to discuss available extension options. For the majority of cases, you will need to submit evidence with a mitigating circumstances request.

For more information relating to mitigating circumstances, click here - [About the Examination Access and Mitigation Committee \(EAMC\) | Cambridge students](#)

To apply for mitigating circumstances, click here - [Guidance notes and application forms | Cambridge students](#)

## Use of AI

Please ensure you are familiar with the following University policies and guidance documents regarding the use of AI:

[Plagiarism and Academic Misconduct: Student Responsibilities](#)

[Plagiarism and Academic Misconduct: Artificial Intelligence](#)

[Guidance: Using Generative AI](#)

[Generative AI and Assessment | Blended Learning Service](#)

## Intellectual Property (IP)

The University's IP policy allows it to fulfil its duty of care to employees and students by requiring disclosure, setting conditions for ownership of intellectual property arising from collaborative environments, and specifying a dispute resolution procedure, to which all parties within the University have access.

The full text of the IP policy is to be found in Chapter XIII, from page 1079 of Statutes and Ordinances of the University of Cambridge:



<https://www.admin.cam.ac.uk/univ/so/pdfs/2022/Cambridge-Statutes-and-Ordinances-2022.pdf>

Students contemplating research projects including data from the 100,000 Genome Project should be aware that separate arrangements may apply regarding IP generated during such projects. More information will be made available when projects are selected and during the information governance training received for accessing the 100,000 genomes project data.

### **General Research Ethics**

The University of Cambridge has a long-standing commitment to the highest standards of ethics in research. Over time, discipline-specific policies and processes have developed to ensure that research at Cambridge not only meets the requirements of legislation and research sponsors, but also satisfies the standards of disciplinary best practice adopted by peers. See [Research Ethics | Research Integrity](#)

For information regarding General Data Protection Regulations (GDPR), see [www.information-compliance.admin.cam.ac.uk/data-protection/general-data-protection-regulation](http://www.information-compliance.admin.cam.ac.uk/data-protection/general-data-protection-regulation) and [www.information-compliance.admin.cam.ac.uk/data-protection/research-participant-data/](http://www.information-compliance.admin.cam.ac.uk/data-protection/research-participant-data/).

Research work requiring ethics approval will not be required for module assignments. However, if you have any questions or concerns, please contact your Course Director.

### **MRes Supervision**

During your MRes you will be allocated a supervisor. Generally, your supervisor will meet you to discuss your progress, provide feedback on assessments and review a draft of your project proposal. Your supervisor will likely be the Course Director.

Your MRes supervisor is also responsible for submitting termly supervision reports that serve as a record of your progress and identify any areas for improvement. You will be notified that a report has been submitted, and you are strongly encouraged to comment on it. Each Michaelmas term, you are invited to complete a self-evaluation report instead of a termly supervision report. The self-evaluation report process is student-led and is an opportunity for you to reflect on your progress to date and highlight any concerns and support needs.

### **MRes Research Project Supervision**

In addition to the MRes supervisor outlined above, all MRes students will have a research project supervisor, strongly aligned with your research goals. It is strongly recommended that you begin a dialogue with the Course Director at an early stage in your studies in regard of this. You will be invited to a meeting with your Course Director to discuss this at the beginning of Michaelmas term. A project supervisor will likely be already working with the data you are analysing or already leading an ongoing research project that you will join.



It is anticipated that they will support you because you will contribute to their research. It is expected that your dissertation supervisor will help you clarify your own ideas and guide your own independent efforts. They will advise on the scope and structure of your dissertation and will help you plan your work.

Your relationship with your dissertation supervisor is a very important part of your course. Your supervisor advises you in relation to your dissertation and engages you in a critical dialogue while you think, research, draft and write. Your dissertation supervisor is responsible for ensuring that you are aware of health and safety issues pertinent to your dissertation work and will advise you about any additional ethical or legal considerations regarding your research.

If you are unable to contact your dissertation supervisor, or do not receive sufficient guidance or encounter any other serious difficulty, you should report this to your MRes supervisor or Course Director in confidence. If you feel unable to approach them, please contact your college tutor or your course administration team.

### MRes Research Project

The research project is a substantial element of the MRes course and its examination. We recommend that you start to think about your dissertation as early as possible. You should begin to formulate your ideas during Michaelmas Term (or earlier) and have discussions with your peers; module leads and your Course Director.

Your final project title must be agreed with your supervisor and then approved by the Course Director on behalf of the Degree Sub Committee. Sometimes, changes to a particular title may be requested, wherever possible consulting the student and supervisor concerned. Once approved, titles cannot be changed without the approval of the Course Director.

Project Proposal Deadline – **Friday 11 December 2026**

Project Submission Deadline – **Friday 27 August 2027**

Project Viva Dates – **First two weeks of September 2027 (exact dates tbc)**

Detailed project guidance will be provided on the MRes Moodle page, including information about style and layout, academic writing, referencing and scholarly conventions, and research ethics.

### College Induction

Colleges may expect or require you to attend induction events. Your college will send you information about induction events and matriculation nearer the time of the course starts. Please note that matriculation is **compulsory**. You cannot be awarded your degree if you have not matriculated. General information about inductions and matriculation can be found at <https://www.cambridgestudents.cam.ac.uk/new-students/arrival>.

## College Tutors

The thirty-one colleges have agreed their own guide for graduate students, which is available at: <http://www.graduate.study.cam.ac.uk/colleges>

This sets out the common core of provision that all Cambridge colleges make for their graduate students, including pastoral support. It explains how this provision works in the college setting, indicating at the same time the diversity that is built into the system. The document also sets out the responsibilities of graduates as members of colleges. College graduate tutors are available for pastoral support and can help with accessing hardship funds. Each college has separate systems in place for graduate tutors, these will be made aware to you by your college during your first weeks in Cambridge.

The Students' Unions' Advice Service also offers free, confidential and independent support to all Cambridge University students, and employs two professional advisors. Contact and service details are available at <http://www.studentadvice.cam.ac.uk>.

## Graduation

Following notification of your award by The Degree Committee of the School of Clinical Medicine in September, at the end of the academic year, your college will arrange your graduation ceremony or congregation (as it is formally known). You should consult the Graduate Secretary and/or Praelector of your college directly for further information.

For a detailed description of degree ceremonies at the University of Cambridge, see: [www.cambridgestudents.cam.ac.uk/your-course/graduation-and-what-next/degree-ceremonies](http://www.cambridgestudents.cam.ac.uk/your-course/graduation-and-what-next/degree-ceremonies)

## Student Concerns and Complaints

We hope that your time studying with will be fulfilling and rewarding, however, should you have any issues or concerns about your course or study we encourage you to talk to the course director, course administration team, and/or your college tutor at the earliest opportunity to ensure the matter can be dealt with promptly. The University also has a number of procedures in place to bring these matters to a productive resolution.

## Appeals, Complaints and Examination Reviews

The University's Office of Student Conduct, Complaints & Appeals handles appeals and complaints relating to all registered students, including those on MPhil degrees. Information on the University's appeals and complaints procedures, including deadlines for submission, for informal and formal academic complaints and, exam reviews is available at [www.studentcomplaints.admin.cam.ac.uk/](http://www.studentcomplaints.admin.cam.ac.uk/)



## The Office for Student Conduct, Complaints and Appeals (OSCCA)

OSCCA is part of the central University and provides procedural information, case handling and oversight of a number of formal student procedures, including student complaints, examination results reviews, student discipline and fitness to study. It is the point of contact for the external ombudsman, the Office of the Independent Adjudicator (OIA), with whom students can raise complaints following the completion of an internal University procedure. Further information at [www.studentcomplaints.admin.cam.ac.uk/](http://www.studentcomplaints.admin.cam.ac.uk/)

## Systems and IT

|                                                   |                                                                                   | Need help?                                                                                                           |
|---------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>VLE</b><br>for all students                    | <a href="https://vle.pace.cam.ac.uk/login/">https://vle.pace.cam.ac.uk/login/</a> | <a href="mailto:elearning@pace.cam.ac.uk">elearning@pace.cam.ac.uk</a><br>or<br>'eLearning Helpdesk' link in the VLE |
| <b>Moodle</b>                                     | <a href="#">Log in to the site   Moodle</a>                                       | <a href="mailto:servicedesk@uis.cam.ac.uk">servicedesk@uis.cam.ac.uk</a>                                             |
| <b>CamSIS</b>                                     | <a href="https://camsis.cam.ac.uk">https://camsis.cam.ac.uk</a>                   | <a href="mailto:servicedesk@uis.cam.ac.uk">servicedesk@uis.cam.ac.uk</a>                                             |
| <b>University IT Helpdesk</b><br>for all students | <a href="https://help.uis.cam.ac.uk/">https://help.uis.cam.ac.uk/</a>             | <a href="mailto:servicedesk@uis.cam.ac.uk">servicedesk@uis.cam.ac.uk</a>                                             |

## Virtual Learning Environment (VLE)

The Genomic Medicine programme is supported and delivered in part through an online VLE, which is central to the course, and a vital study resource and communication channel. It is also the place where assignments are submitted, and feedback is received.

The course material, assessment and timetables are all delivered over the VLE. Presentations given by lecturers and speakers will be uploaded to the VLE. Not every speaker is willing to share their lecture material, and some may contain unpublished or proprietary information. It is strongly suggested that you take extensive notes unless you are advised otherwise.

On the VLE, you will also have access to many other online resources to enable you to get the most from your studies.

If you encounter any problems, please contact the VLE Helpdesk: [elearning@pace.cam.ac.uk](mailto:elearning@pace.cam.ac.uk)

## Moodle

Moodle is the Central University online workspace. There is a dedicated space for MRes students containing information about your research project and supervisions. You will need your University Account (see page 12) formerly known as your RAVEN ID to log in to Moodle.



## Getting Started – Links, Guides and Training for IT and Computing

For IT guides and useful links to further information this is the best place to start:

<https://help.uis.cam.ac.uk/new-starters/it-for-students>

There are links for setting up Eduroam internet access and many more IT related items, including information about the various passwords and how to collect and set them up.

The University IT support service desk is at <https://help.uis.cam.ac.uk/>

Tel: 01223 762999 Email: [service-desk@uis.cam.ac.uk](mailto:service-desk@uis.cam.ac.uk)

Students are expected to take personal responsibility for ensuring that they are familiar with the software packages that they will need during the course of their studies. The University Computing Service offers a wide variety of instructor-led and self-taught training courses for beginners and advanced users at <https://www.training.cam.ac.uk/ucs/>

Inexpensive laser printing, photocopying and other services are also available at the Cambridge Student Union, see: <https://www.cambridgesu.co.uk/aboutus/services/printshop/>  
MSt students may also be able to access printing and IT support through their colleges.

Please ensure you back up your course files and in particular assignment and dissertation files, as technological problems do not constitute good grounds for seeking extensions to assignments or your dissertation.

### Software

The University's Information Services has negotiated a number of arrangements which provide free or heavily discounted software for Cambridge students, see: <https://www.uis.cam.ac.uk/>

### Wi-Fi

Wi-fi facilities - UniofCam and Eduroam - are available throughout the University. You will require a University Account password to gain access to UniofCam and a 16-digit code for Eduroam. Eduroam provides internet access in a wide and increasing number of universities around the world.

## University of Cambridge Resources

Students have access to a wide range of facilities, discounts, training, and services to ensure your time with us is as enjoyable and productive as possible. Some of these include:

### Talks, Seminars and Conferences

There are many talks and seminars that may be particularly relevant to students of Genomic Medicine.

These seminars provide a wide perspective on clinical research and practice and are usually delivered by outstanding speakers. The following series may be of particular interest:

[Clinical School Seminar Series](#)

[CRUK seminar series](#)

**Genomic Medicine Newsletter**

Students on the Genomic Medicine programme are subscribed to a weekly course newsletter which will highlight course news as well as many additional opportunities for Genomic Medicine specific learning and engagement.

**Cambridge University Libraries**

There are numerous university libraries which you can access, some requiring the University card. The University Library (on West Road), the Medical Library (in the Clinical School building) and your own college library will be particularly useful. The Medical Library in the Clinical School affords a wide spectrum of books and journals.

**The University Library**

The University Library (UL) on the outskirts of the city centre is likely to be your main source for research and is affiliated with over 20 University departmental and faculty libraries in Cambridge. It is a legal deposit library and a repository of official documents of the UK government and of many international organisations. To gain access to the library, you will need to have your University Card as this also serves as your University Library card and is a prerequisite for registration at some faculty and departmental libraries. Please see [www.lib.cam.ac.uk/](http://www.lib.cam.ac.uk/) for further information.

iDiscover is the University Library's catalogue system. This enables students to simultaneously search across the print items held in the University's libraries and the numerous electronic resources (ebooks, ejournals and databases) to which the University subscribes, together with the open access items in the University's repository. See <http://idiscover.lib.cam.ac.uk> for further information and to manage your loans, requests, fines and saved search results.

Cambridge University Library participates in the Society of College, National and University Libraries (SCONUL) Vacation Access scheme. This enables undergraduate and postgraduate students following taught courses at the University of Cambridge to have reference use of other higher education libraries free of charge. If you request admittance to another library, you should be prepared to present proof of your current membership of Cambridge University (i.e. your university card). Please check the SCONUL information for each library you wish to access [www.sconul.ac.uk/page/about-sconul](http://www.sconul.ac.uk/page/about-sconul)

**The Medical Library**

The Medical Library collections support all aspects of medicine and healthcare. The book collection, over 50,000 volumes, contains core textbooks and specialty titles many of which are available electronically and in print. The journal collection is increasingly weighted towards electronic access, with over 104,000 e- titles. The library also facilitates access to the NHS Core Content collection of electronic resources. Please see the [Medical School Library](#) for further information.

The Medical Library provides several computers for student use. These are equipped with common Office applications, the SPSS statistical package, and the EndNote referencing



tool. Printing facilities and a scanner are also available. Students can also bring their own devices.

Library training courses, including Medical Library specific courses, can be found at <https://training.cam.ac.uk/cul/event-timetable>

### **Cambridge Students Union**

In July 2020 Cambridge SU replaced the Cambridge University Student Union and the Graduate Union, becoming one central, single union for all students. Each Cambridge MPhil student is automatically a member of the Cambridge SU, which ensures that all students, regardless of level of study, are represented, supported and have their voice heard across the University of Cambridge. The Student Union provides a free independent Advice Service, campaigns for marginalised groups and provides services such as dissertation binding and loan of gowns. Cambridge SU is located at 17 Mill Lane, Cambridge, CB2 1RX. See <https://www.cambridgesu.co.uk/>

### **Cambridge University Press**

You are entitled to 20% discount on books published by Cambridge University Press (CUP) purchased at the Press Bookshop, 1 Trinity Street, Cambridge. You will need to show your university card as evidence of study.

### **Cambridge University Botanic Garden**

You can gain free entry to the University Botanic Garden by presenting your university card.

### **The University Sports Centre**

The University's Sports Centre is in West Cambridge on the outskirts of the city centre. Anyone can become a member and use the facilities and there are a number of competitive membership and pay-as-you-go options. You will need to take your University Card. See [www.sport.cam.ac.uk/](http://www.sport.cam.ac.uk/)

## **Continuation to PhD**

The MRes in Genomic Medicine provides an excellent transition to full-scale PhD research. It introduces and develops research skills and specialist knowledge. The MRes is also an invaluable experience in preparing and working on a research thesis. There is no automatic progression to a PhD. Those interested in applying for a PhD are encouraged to apply early to the university to ensure they are considered in funding schemes.

A workshop will be held at the end of the Michaelmas term to prepare those applying for PhD interviews. Please let Dr Hearn know during Michaelmas if you would like to take advantage of this session.

Further information about doctoral studies in the University of Cambridge is available on the [Graduate School of Life Sciences](#) website. This website contains a database of potential

projects and supervisors in the Clinical School. You will hear more about the research of members of the Clinical School during the MPhil.

## Travel Around Cambridge

### Biomedical Campus

There is a bus service (U) that runs frequently during the working day (every 15 minutes) between Madingley Road Park and Ride, University West Cambridge Site and the Biomedical Campus. Prices are discounted to £1 each way for university members. You must show your university card when boarding.

### The West Hub

In line with the University's commitment to sustainability, the West Hub does not have any dedicated parking. The nearest public parking is at the Madingley Road Park and Ride, which is just a short walk from the building.

We strongly encourage visitors to use public transport, cycle, or arrive on foot where possible. From the city centre, the West Hub is a:

- 33-minute walk via the scenic Coton footpath
- 10-minute cycle ride
- Short ride on bus routes 4, 8, U, X3, or the Madingley Road Park & Ride service

Cycle parking is available in multiple areas around the building, including the South Plaza off JJ Thomson Avenue. Showers and lockers can be found on the ground floor next to the canteen. Please note, for security reasons, lockers are cleared out each night, so remember to take your belongings with you at the end of the day.



## GM1 – Fundamentals of Human Genetics and Genomics

### Module Lead – Dr Gemma Chandratillake (Cambridge University Hospitals)

This module contains approximately 30 hours of face-to-face teaching. This introductory module aims to provide the student with an introduction to the key areas of genomics, human genetics and genetic variation. It will prepare participants to understand disease genetics and how genomic medicine can be utilised to elucidate disease mechanisms and biology. In addition, this module will also cover the fundamentals of information governance in the context of genomic medicine and its applications providing underpinning knowledge for later modules in bioinformatics and statistics. This module will serve as a foundation for those wishing to advance their careers within the NHS in genomic medicine. This module provides clear understanding of the structure and variations in genetic material. Covering basic genetics and genomics, it will prepare students to understand the role of genetics in disease and how genomic information can be utilised to elucidate disease mechanisms and biology.

### Content

In GM1 you will learn about:

- Structure and function of nucleic acids and chromosomes
- Architecture and organisation of the human genome and genetic variation within it
- DNA sequence variation, type and frequency e.g. single nucleotide variants, small insertions and deletions, copy number variation, rearrangements and tandem repeats
- How variation arises and its extent in populations (e.g. HapMap)
- Gene regulation: enhancers, promoters, transcription factors, silencers
- Epigenetics and imprinting
- Mutational mechanisms: how different types of DNA variants affect gene function or expression to cause disease; correlation of genotype with phenotype
- Concepts of penetrance, expressivity, heterogeneity and pleiotropy
- Modes of inheritance for clinical manifestation of human variation
- Types of genetic testing (prenatal, carrier, predictive, diagnostic, cascade, screening, etc)
- Germline, mitochondrial and somatic mutations
- Chromosomal abnormalities and disease
- Mechanisms of Mendelian disorders
- Interpretation of genetic variation
- Basic Bayesian approach to calculating disease probabilities
- Overview of key ethical, legal and social implications of genomics
- Introduction to some of the principles underlying application of genomic data in clinical practice
- Legislation, Codes of Practice, Caldicott Guardian and Information Commissioner
- Patient identifiable data and information, relationship between data and information
- Information system risks to patient safety, electronic and paper copies, safe havens, encryption, secondary uses of data, audit and research
- Secure information exchange between professionals



- Sharing and communication with patients and careers, consent
- Handling requests for information about patients/clients.

### Learning Outcomes

By then end of GM1 you will be able to:

- Discuss the human genome structure and the properties of DNA
- Critique genome architecture and its variation across human populations
- Critically evaluate the regulation of gene expression, transcription and translation
- Appraise and interpret variation in genome structure and sequence in the context of physiological function and disease
- Discuss and analyse epigenetic modifications and imprinting and its role in disease
- Correlate genetic markers to phenotype and interpret output of association studies both for dichotomous and quantitative traits
- Discuss different disease mechanisms and modes of inheritance
- Apply concepts of inheritance and calculate genetic risks for Mendelian conditions
- Discuss and justify the ethical and governance frameworks in place within the NHS and how they apply to medical genomics including patient safety, data sharing and confidentiality
- Identify the range, purposes, benefits and potential risks of sharing, integrating and aggregating clinical data and information.
- Describe and evaluate the purpose, structures, use and storage of health records.



## GM2 – Research and Statistical Skills

**Module Leads: Dr Tim Hearn (University of Cambridge) and Dr Leonardo Bottolo (University of Cambridge)**

This module aims to provide and equip students with the knowledge of statistics and computational tools needed to independently complete genomic medicine research. This module complements the knowledge from the wider course to ensure all students are competent in planning research, and can perform appropriate statistical analysis in R. The module is designed to prepare students for research projects and presents material in this context. This module serves as an informatics foundation for students learning bioinformatics and statistical languages. In particular, the module introduces students to the Genomics England Research Environment, and provides both self-directed, and supervised learning of the Unix command line environment and the R statistics language. This module provides an excellent foundation for performing a research project on the Genomics England 100,000 genomes data, and for students considering undertaking a PhD in future years.

### Course content

*In GM2 students will learn about:*

- How to write a research grant proposal
- Up to date computational and experimental tools for genomic medicine research projects
- The Genomics England research environment and how to perform research on the 100,000 genomes dataset.
- How to use the Unix command line and Galaxy to perform basic genomic analysis
- Fundamental concepts in statistics offering students a foundation and framework for understanding more complex methods.
- Implementation of the statistical methods illustrated during the course using the R environment.
- Basic concepts in using R
- Basic statistical and mathematical concepts including correlation, regression, model selection, and generalized linear models
- Exploration of Bayesian statistics and tools applied for haplotype estimation and Machine Learning tools for non-linear regression will also be explored

## Learning Outcomes

*By the end of GM2 students will be able to:*

- Demonstrate a rounded knowledge of the classic and modern tools and technologies used in research projects. Emphasis will be placed on the overlap between modern and traditional approaches and the type of data they might be presented within a research project
- Understand the conceptual basis for statistical tests, and be aware of the various tools that they can implement when exploring either genomic or expression data
- Choose the correct statistical test based on the data source and experimental design and implement the test in R
- Have a thorough understanding of the types of computational tool available for use in research projects, and the best platform to use them on (Galaxy, command line, HPC etc.)
- Have a good understanding of how to access the central University Linux cluster from an external computer, in preparation for accessing remote High Performance Computing services.





## GM5 – Bioinformatics, Interpretation and Data Quality Assurance in Genomic Analysis

**Module Leads: Dr Tim Hearn (University of Cambridge) and Dr Kenneth Langlands (University of Cambridge)**

The aim of this module is to equip students with the skills to analyse and interpret sequencing data, including variant identification, annotation and filtering using appropriate computational and statistical methods. Students will also critically assess the challenges of managing large genomic datasets and the role of databases in variant annotation. This module covers the fundamental principles of bioinformatics with a focus on its impact in both clinical genomics and in research. Students will gain hands-on experience in analysing sequencing data, including quality assessment, alignment to reference genomes, variant calling, and annotation. They will also learn to apply computational skills and statistical methods to handle large genomic datasets and critically evaluate the use of different databases for variant annotation. Through a combination of theoretical sessions and practical exercises, students will develop the essential skills to analyse and interpret genomic data, preparing them to address current and future challenges in managing and utilizing large-scale genomic information in clinical and research settings.

### Content

*In GM5 students will learn about:*

- bioinformatics principles and their application in clinical genomics
- techniques for assessing the quality of raw sequencing data
- common quality metrics (e.g., base quality scores, read length distribution, coverage) and the use of appropriate tools and software
- genome alignment and variant calling:
  - aligning sequencing data to reference genomes using tools like BWA and Bowtie.
  - variant calling methods: single nucleotide polymorphisms (SNPs), insertions and deletions (indels)
  - best practice in variant calling and filtering strategies to identify clinically- relevant variants
- variant annotation and interpretation:
  - annotating variants using databases like dbSNP, ClinVar, and Ensembl.
  - tools for variant annotation
  - interpretation of annotated variants and their clinical significance.
- computational and statistical methods in genomic data analysis:
  - basic computational skills for handling large datasets (e.g., command-line tools, scripting).
  - statistical methods for variant analysis and interpretation.
  - data normalization, variant frequency calculations and significance testing
- databases and resources for genomic data:
  - overview of major genomic databases and their use in variant annotation (e.g., 1000 Genomes, ExAC, gnomAD)



- evaluating the reliability and relevance of different databases
- integrating data from multiple sources for comprehensive variant interpretation.
- challenges in managing large genomic datasets:
  - current and future challenges in genomic data management (e.g., storage, processing power, data security).
  - techniques for efficient data handling and processing (e.g., cloud computing, parallel processing).
  - strategies for maintaining data integrity and privacy in large-scale genomic research.
  - ethical, legal, and social issues in the use of genomic data.
- future trends in genomic data analysis:
  - emerging technologies and methodologies in genomic data analysis.
  - the role of artificial intelligence and machine learning in variant interpretation.

### Learning Outcomes

*By the end of GM5 students will be able to:*

- analyse the quality of sequencing data, align to a reference genome, call and annotate variants, and apply filtering strategies to identify variants in sequencing data
- appraise the use of different databases in variant annotation
- apply relevant basic computational skills and statistical methods to handle and analyse genomic data.
- examine the current and future challenges of managing and manipulating large genomic data sets

